

Shahjalal University of Science and Technology, Sylhet

Department of Electrical and Electronic Engineering



Project Proposal

IoT-Based Health Monitoring System Using ESP32

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Contents

1	Introduction	2
2	Problem Statement	2
3	Objectives of the Project	2
4	System Overview	2
5	Block Diagram of the Proposed System	3
6	Circuit Diagram	4
7	Printed Circuit Board (PCB) Design	5
8	Hardware Components	6
9	Methodology	6
10	Estimated Project Cost	7
11	Patient Benefit and Application	7
12	Conclusion	8

1. Introduction

The rapid advancement of Internet of Things (IoT) technology has enabled modern healthcare systems to provide continuous and remote monitoring of patients. Elderly, chronically ill, and post-operative patients often require continuous health observation, which is difficult to ensure using traditional hospital-based systems. This issue is more prominent in developing countries like Bangladesh, especially in rural and isolated areas.

This project proposes a portable IoT-based health monitoring system using ESP32 that can monitor vital physiological parameters and transmit real-time data to caregivers or medical professionals.

2. Problem Statement

Traditional healthcare monitoring systems face several challenges:

- Lack of continuous monitoring outside hospitals
- High healthcare costs
- Limited accessibility in rural areas
- Delayed emergency response

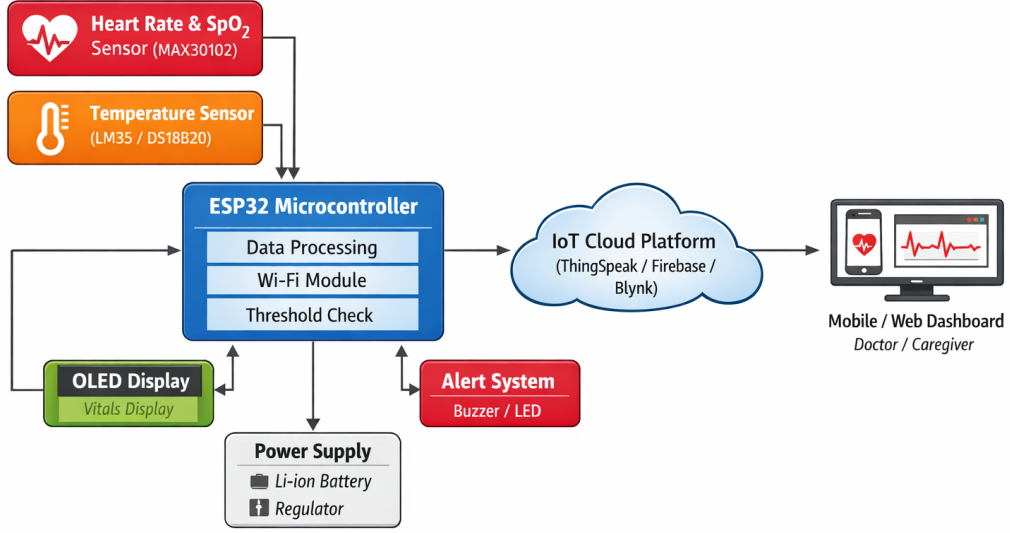
3. Objectives of the Project

- Design a portable health monitoring system using ESP32
- Measure heart rate, SpO₂, and body temperature
- Transmit real-time data using Wi-Fi
- Store and visualize data on an IoT cloud platform
- Generate alerts during abnormal health conditions

4. System Overview

The system consists of biomedical sensors connected to an ESP32 microcontroller. Sensor data is processed and transmitted wirelessly to an IoT cloud platform for remote monitoring.

5. Block Diagram of the Proposed System



Block Diagram of the Proposed System

Figure 1: Block Diagram of the IoT-Based Health Monitoring System

The block diagram illustrates the complete architecture of the IoT-based health monitoring system. The MAX30102 sensor measures heart rate and blood oxygen saturation (SpO₂), while the temperature sensor continuously monitors body temperature. These sensors interface with the ESP32 microcontroller through I2C and analog communication protocols. The ESP32 processes the acquired physiological data and displays real-time health parameters on the OLED screen for local monitoring. Simultaneously, the processed data is transmitted wirelessly via Wi-Fi to a cloud platform such as ThingSpeak or Blynk, enabling remote access for healthcare professionals and family members. The system incorporates an alert mechanism using a buzzer and LED indicator that activates when vital parameters exceed predefined threshold values, ensuring immediate attention during critical health conditions. The entire system operates on a portable Li-ion battery with regulated power supply, making it suitable for continuous monitoring in home-based and ambulatory care settings.

6. Circuit Diagram

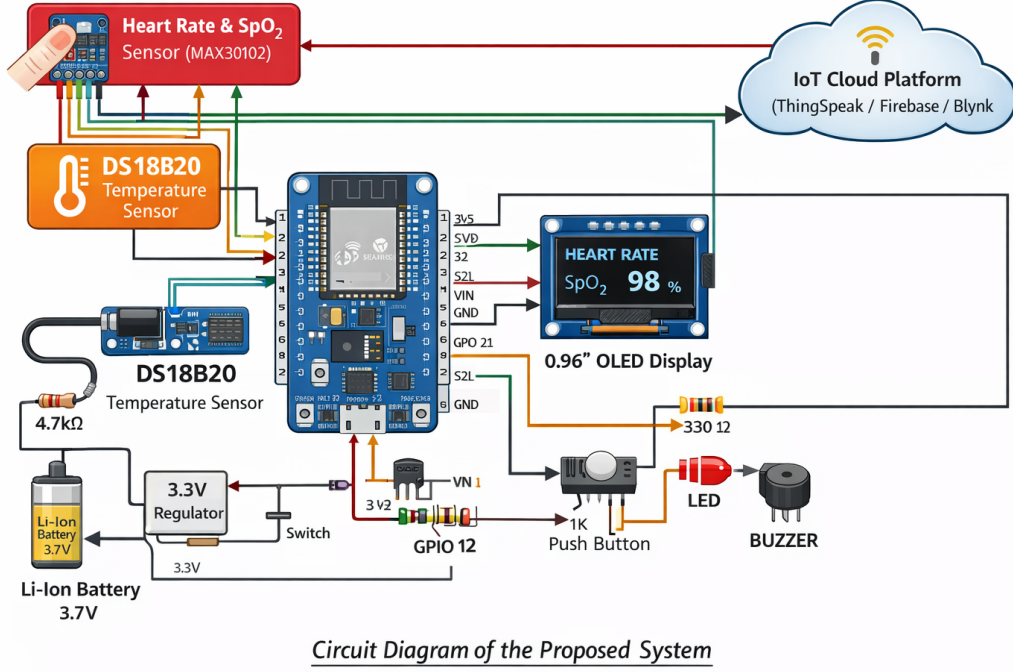


Figure 2: Circuit Diagram of the Proposed System

The circuit diagram presents the detailed electrical connections and interfacing architecture of the health monitoring system. The ESP32 microcontroller serves as the central processing unit, operating at 3.3V logic level. The MAX30102 pulse oximetry sensor is connected to the ESP32 via the I2C communication protocol, utilizing the SDA (GPIO21) and SCL (GPIO22) pins for bidirectional data transfer. The temperature sensor (DS18B20 or LM35) interfaces through an analog input pin (ADC1) with appropriate pull-up resistors to ensure accurate temperature readings. The 0.96-inch OLED display is also connected via I2C, sharing the same communication bus with the MAX30102 sensor. The alert system consists of an active buzzer and LED connected to digital output pins through current-limiting resistors and NPN transistors for proper current handling. A 3.7V Li-ion battery powers the entire system through a voltage regulator module (AMS1117-3.3V or similar) that provides stable 3.3V supply to all components. The circuit incorporates decoupling capacitors near the power pins of each IC to minimize noise and ensure stable operation during wireless data transmission.

7. Printed Circuit Board (PCB) Design

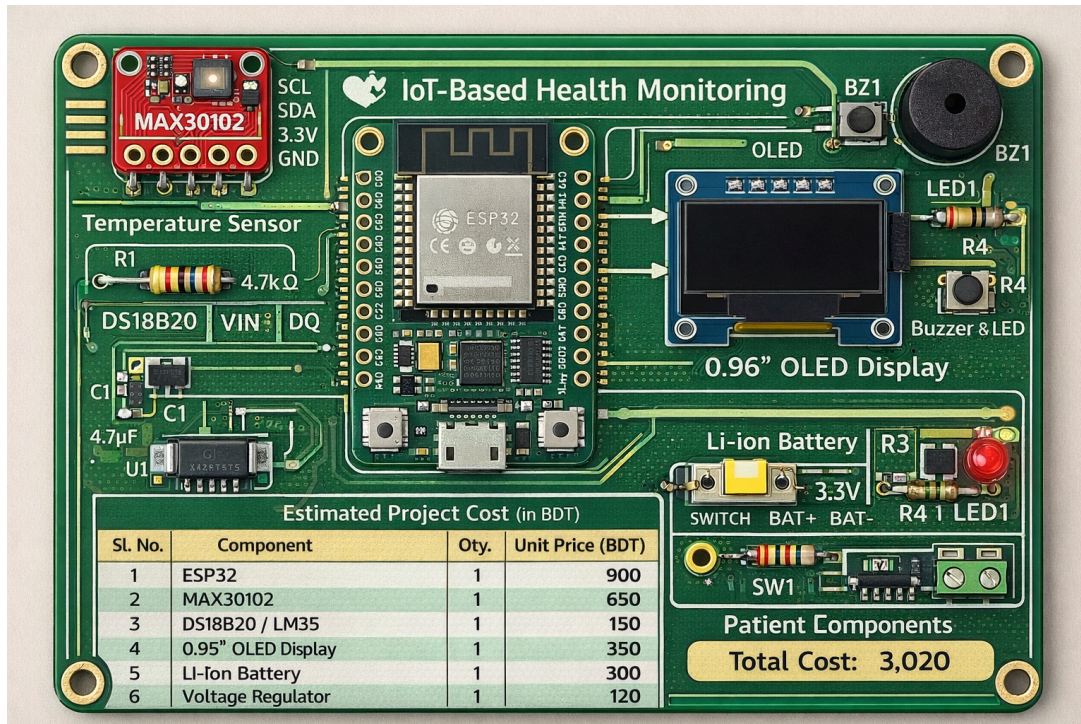


Figure 3: PCB Layout of the IoT-Based Health Monitoring System

The PCB design provides a compact and professional implementation of the health monitoring system, eliminating the need for breadboard connections and ensuring reliable long-term operation. The PCB layout is designed as a double-sided board with dimensions of approximately 70mm $\times$ 50mm to maintain portability. The top layer accommodates the ESP32 module, sensor breakout boards, and OLED display, while the bottom layer houses the voltage regulator circuit and power management components. Dedicated ground planes are incorporated on both layers to minimize electromagnetic interference and provide a stable reference voltage for analog sensor readings. The I2C bus traces are routed with equal lengths to prevent signal skew and maintain communication integrity. Proper trace width calculations ensure adequate current handling capacity for power lines, with wider traces (1.5mm) used for the battery supply and regulated 3.3V rails. Through-hole vias connect the top and bottom layers at strategic points for ground connections and signal routing. The PCB includes mounting holes for secure installation in a protective enclosure, and connector pads are provided for easy battery replacement and sensor attachment. This professional PCB implementation enhances system reliability, reduces noise susceptibility, and provides a production-ready design suitable for practical deployment.

8. Hardware Components

- ESP32 Microcontroller
- MAX30102 Heart Rate and SpO₂ Sensor
- LM35 / DS18B20 Temperature Sensor
- OLED Display
- Li-ion Battery
- Voltage Regulator
- Buzzer and LED

9. Methodology

1. Sensor interfacing with ESP32
2. Data acquisition and processing
3. Wireless data transmission to cloud
4. Cloud-based data visualization
5. Alert generation

10. Estimated Project Cost

Table 1: Estimated Project Cost in Bangladeshi Taka (BDT)

Sl. No.	Component	Qty.	Unit Price (BDT)	Total (BDT)
1	ESP32 Development Board	1	900	900
2	MAX30102 Sensor	1	650	650
3	Temperature Sensor	1	150	150
4	OLED Display	1	350	350
5	Li-ion Battery	1	300	300
6	Voltage Regulator Module	1	120	120
7	Buzzer	1	50	50
8	LED and Resistors	1 set	50	50
9	Wires and Breadboard	1 set	200	200
10	PCB / Project Box	1	250	250
Total Estimated Cost				3,020

11. Patient Benefit and Application

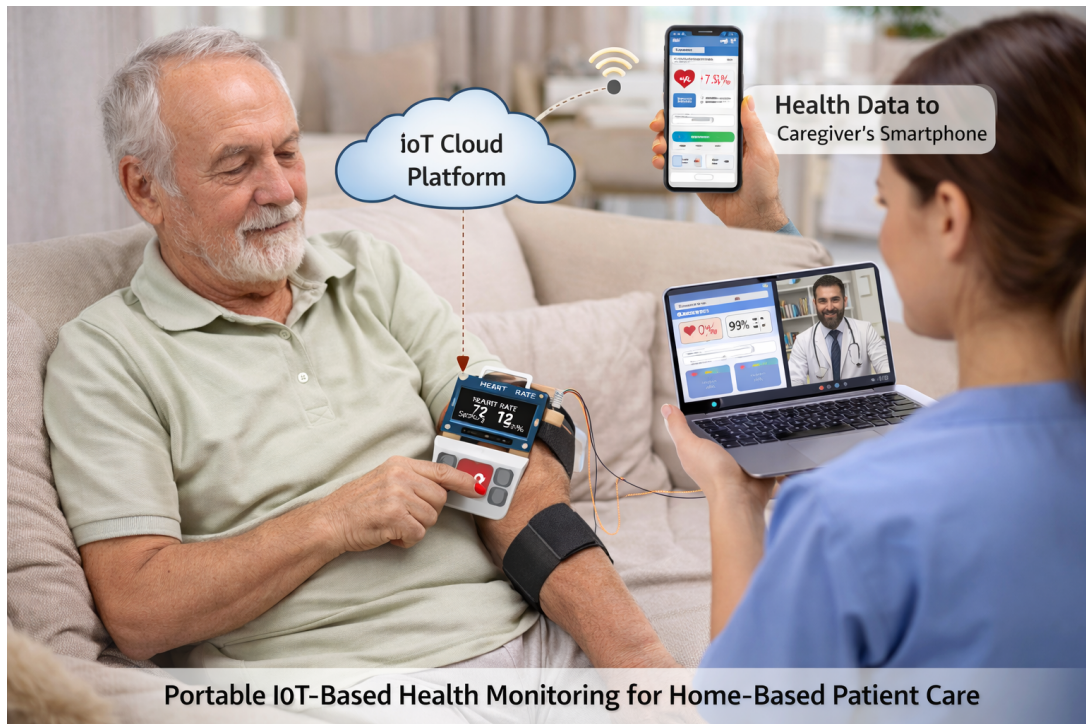


Figure 4: Application of the Proposed System for Home-Based Patient Care

The system enables continuous health monitoring at home, reducing hospital visits and improving patient safety.

12. Conclusion

This project presents a portable IoT-based health monitoring framework using ESP32 that provides a low-cost and efficient solution for isolated and home-based patient care.